

Brief Report

Does lithium protect against dementia?

Kessing LV, Forman JL, Andersen PK. Does lithium protect against dementia?

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Objective: To investigate whether treatment with lithium in patients with mania or bipolar disorder is associated with a decreased rate of subsequent dementia.

Methods: Linkage of register data on prescribed lithium in all patients discharged from psychiatric health care service with a diagnosis of mania or bipolar disorder and subsequent diagnoses of dementia in Denmark during a period from 1995 to 2005.

Results: A total of 4,856 patients with a diagnosis of a manic or mixed episode or bipolar disorder at their first psychiatric contact were included in the study. Among these patients, 2,449 were exposed to lithium (50.4%), 1,781 to anticonvulsants (36.7%), 4,280 to antidepressants (88.1%), and 3,901 to antipsychotics (80.3%) during the study period. A total of 216 patients received a diagnosis of dementia during follow-up (103.6/10,000 person-years). During the period following the second prescription of lithium, the rate of dementia was decreased compared to the period following the first prescription. In contrast, the rates of dementia during multiple prescription periods with anticonvulsants, antidepressants, or antipsychotics, respectively, were not significantly decreased compared to the rate of dementia during the period with one prescription of these drugs.

Conclusions: Continued treatment with lithium was associated with a reduced rate of dementia in patients with bipolar disorder in contrast to continued treatment with anticonvulsants, antidepressants, or antipsychotics. Methodological reasons for these findings cannot be excluded due to the nonrandomized nature of the data.

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Studies suggest that bipolar disorder is associated with an increased risk of developing dementia (1, 2) and that the risk increases with every new affective episode (3). On the other hand, it has also been suggested that lithium may have neuroprotective abilities and may reduce the risk of developing dementia, as lithium inhibits glycogen synthase kinase-3, which is a key enzyme in the metabolism

LVK had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. LVK has been a consultant for Bristol-Myers Squibb, Eli Lilly & Co., Lundbeck, AstraZeneca, Pfizer, Wyeth, and Servier. However, none of the authors of this paper have any commercial associations that might pose a conflict of interest in connection with this manuscript.

of amyloid precursor protein and in the phosphorylation of tau protein involved in the pathogenesis of Alzheimer's disease (4–7). The association between lithium consumption and dementia has only been investigated in three human studies. We recently published results from a cohort study linking registers of all prescribed lithium and diagnoses of dementia in Denmark during a period from 1995 to 2005 (8). The study included a total of 16,238 persons who purchased lithium at least once and 1,487,177 persons from the general population who did not purchase lithium. As expected, persons who purchased lithium once had an increased rate of dementia compared to persons unexposed to lithium [rate ratio = 1.47; 95% confidence interval: 1.22–1.76]. Nevertheless,

continued lithium treatment was associated with a reduced rate of dementia similar to the rate for the general population. In contrast, continued use of anticonvulsants was not associated with decreased dementia. It was, however, a limitation of the study that indications for lithium were not known, i.e., it was not known whether patients using lithium suffered from mania/bipolar disorder, unipolar depressive disorder, or other conditions. Further, two small studies including 87 patients (9) and 114 patients (10), respectively, on lithium therapy have found diverging results; one study suggested an increased risk of dementia with increasing numbers of lithium prescriptions (9) in opposition to the other study, which found that patients on chronic lithium therapy had decreased prevalence of Alzheimer's disease compared to patients without recent lithium therapy (10).

The aim of the present study was to investigate whether continued treatment with lithium in patients with mania or bipolar disorder is associated with reduced risk of developing dementia by linking Danish nationwide registers of psychiatric hospital contacts, prescribed lithium, and diagnoses of dementia given at psychiatric hospital inpatient or outpatient settings in Denmark. Further, it was the aim to investigate, in the same population of bipolar patients, whether continued treatment with anti-convulsants, antidepressants, and antipsychotics, respectively, was eventually associated with a decreased rate of developing dementia.

Methods

The registers

Data were obtained by linking Danish population-based registers using the unique personal identification numbers that are assigned to all 5.3 million persons living in Denmark, thus ensuring accurate linkage of information between registers, irrespective of changes in name, etc. (11). In this way, the Medicinal Product Statistics (12) was linked with the Danish Medical Register on Vital Statistics (13), the Danish National Hospital Register (DNHR) (14), and the Danish Psychiatric Central Register (DPCR) (15).

- (i) The Medicinal Product Statistics contains data on all prescribed medication purchased at pharmacies from 1 January 1995 onwards (12). In Denmark, all medications prescribed by doctors, such as lithium, are purchased at pharmacies only. Information on indication for the drug is not available.
- (ii) The Danish Medical Register on Vital Statistics (13) contains data on deaths.

- (iii) The DNHR (14) contains data on all patients treated at all somatic hospitals as in- or outpatients in Denmark from 1 January 1977 onwards as a part of the official Danish health survey (16).
- (iv) Likewise, all psychiatric admissions have been registered in a nationwide register, the DPCR (15), from 1 April 1970 onwards. In Denmark, the International Classification of Diseases (ICD)-8 was used prior to 1994, and since 1 January 1994, the ICD-10 has been in use in both registers (16).

The study period was from 1 January 1995 to 31 December 2005, although the period prior to 1995 was used to exclude patients with a prior diagnosis of dementia or schizophrenia (back to 1977 in the DNHR, and back to 1970 in the DPCR).

The sample

The sample was defined as all patients who, at their first psychiatric hospital contact as inpatients or outpatients during the study period from 1995 to 2005 (the index contact), inclusively, were given a discharge main diagnosis of a manic or mixed episode or bipolar disorder (ICD-10 codes: F30–31.9 and 38.0) and who purchased lithium, anticonvulsants, antidepressants, or antipsychotics at least once during the study period. Patients who prior to this diagnosis were given a diagnosis of dementia (see ICD-10 codes below) or schizophrenia (ICD-8: 295.09–295.99; ICD-10: F20–29) were excluded.

Purchase of lithium, anticonvulsants, antidepressants, and antipsychotics was identified in the Medicinal Product Statistics during the study period from 1 January 1995 to 31 December 2005.

Main and auxiliary diagnoses of dementia given following a hospitalization or outpatient contact were identified in the DNHR (ICD-10 codes: G30.0–G30.9) and the DPCR (ICD-10 codes: F.00–F00.9 and F0.1–F01.9) during the period from 1 January 1995 to 31 December 2005.

Statistical analysis

Poisson regression analyses were conducted with a diagnosis of dementia as outcome and with the number of prescriptions of lithium, anticonvulsants, antidepressants, and antipsychotics, respectively, as the variables of interest and censoring at the time of a diagnosis of schizophrenia, death, or at the end of the study period (31 December 2005). Poisson regression models (17) are standard multiple regression models for incidence rates where numbers of diagnoses with dementia and person-years at risk

are computed and analyzed in subgroups given by covariates, which may be time dependent. Confounding by indication may occur if the effect of the various drugs on the rate of dementia was compared directly, since the four types of drugs are given to patients with different types and severity of bipolar disorder. Thus, we chose to compare periods with multiple prescriptions with the period with one prescription within each of the four types of drugs. In our previous studies on the associations between lithium and dementia (8), antidepressants and dementia (18), and lithium and suicide (19), respectively, we have used the same kinds of analyses to reduce confounding by indication. Time-dependent exposure variables were defined for each subject in turn on a day-to-day basis as the total number of prescriptions of the given drug redeemed up to and including the date in consideration. Risk time was estimated from the age of 40 years to the age of 100 years.

Sex, age, and calendar period in one-year periods were included in the model as covariates. Additionally, purchasing drugs other than the drug of interest was included in the model as time-dependent variables (e.g., in the analysis of the association between lithium and dementia, the model was adjusted for any purchase of anticonvulsants, antidepressants, or antipsychotics). SAS statistical software, version 9.1.3, was used.

Results

During the study period from 1 January 1995 to 31 December 2005, a total of 5,011 patients were given a diagnosis of a manic or mixed episode or bipolar disorder at their first psychiatric contact. These patients were older than 40 years at the index contact and had not had a prior diagnosis of dementia or schizophrenia. Among these patients, 4,856 (100%) were exposed to any of the drugs of interest at least once during the study period: 2,449 (50.4%) were exposed to lithium, 1,781 (36.7%) to anticonvulsants, 4,280 (88.1%) to antidepressants, and 3,901 (80.3%) to antipsychotics. A total of 155 patients (3.0% of the total sample) were unexposed to any of these drugs and were excluded from analyses, as the diagnosis for these individuals was considered less valid since they got no medication for bipolar disorder. Table 1 presents characteristics of patients exposed to the various drugs according to sex, age, number of persons with a diagnosis of dementia, and censoring. A total of 216 patients had a diagnosis of dementia of any kind, corresponding to 4.5% of the total sample of 4,856 patients who were exposed to any of the four types of drugs.

Table 2 presents the various drug combinations for the four types of drugs. Patients were rarely exposed to only one type of drug during the study

Table 1. Characteristics of patients with a diagnosis of mania or bipolar disorder who were exposed to lithium, antidepressants, anticonvulsants or antipsychotics, respectively, according to sex, age, number of persons with a diagnosis of dementia, and censoring

	Lithium	Anticonvulsants	Antidepressants	Antipsychotics	Any exposure
N at end of follow-up (%)	2,449 (50.4)	1,781 (36.7)	4,280 (88.1)	3,901 (80.3)	4,856 (100)
Total, person-years	11,767	7,465	18,441	16,789	20,846
Sex, person-years					
Male	4,798	2,985	7,160	6,666	8,417
Female (%)	6,969 (59.2)	4,480 (59.2)	11,281 (61.2)	10,123 (60.3)	12,430 (59.6)
Age at inclusion, median (quartiles)	52.5 (45.1–61.6)	51.8 (44.4–60.4)	53.9 (46.1–64.2)	54.2 (46.1–64.4)	54.1 (46.1–64.5)
No. of patients with dementia (all inclusive)					
Total	84	50	186	185	216
Per 10,000 person-years	71.4	67.0	100.9	110.2	103.6
No. of patients with Alzheimer's dementia					
Total	19	7	31	36	40
Per 10,000 person-years	16.1	9.4	16.8	21.4	19.2
No. of patients with other dementia					
Total	65	43	155	149	176
Per 10,000 person-years	55.2	57.6	84.1	88.8	84.4
Censoring, n (%)					
Schizophrenia	59 (2.4)	50 (2.8)	145 (3.4)	158 (4.1)	178 (3.7)
Death	297 (12.1)	204 (4.8)	599 (14.0)	565 (14.5)	692 (13.8)
End of study	2,009 (82.0)	1,477 (82.9)	3,350 (78.3)	2,993 (76.7)	3,770 (77.6)

Table 2. Pharmacological treatment of patients with a diagnosis of mania or bipolar disorder at end of study period

Drug(s)	n (%)
Lithium	35 (0.7)
Anticonvulsants	12 (0.2)
Antidepressants	397 (8.2)
Antipsychotics	251 (5.2)
Lithium, anticonvulsants	8 (0.2)
Lithium, antidepressants	276 (5.7)
Lithium, antipsychotics	154 (3.2)
Lithium, anticonvulsants, antidepressants	97 (2.0)
Lithium, anticonvulsants, antipsychotics	58 (1.2)
Lithium, antidepressants, antipsychotics	1,006 (20.7)
Lithium, anticonvulsants, antipsychotics, antidepressants	815 (16.8)
Anticonvulsants, antidepressants	130 (2.7)
Anticonvulsants, antipsychotics	58 (1.2)
Anticonvulsants, antidepressants, antipsychotics	603 (12.4)
Antidepressants, antipsychotics	956 (19.7)
Total	4,856 (100)

period; e.g., 0.7% of the population was exposed to lithium only. The most prevalent combinations of drugs were: (i) lithium, antidepressants, and antipsychotics (20.7%); (ii) antidepressants and antipsychotics (19.7%); (iii) all four types of drugs (16.8%); and (iv) anticonvulsants, antidepressants, and antipsychotics (12.4%).

Table 3 presents the rate of dementia related to number of prescriptions (1, 2–4, 5–9, 10–19, ≥ 20) of the various types of drugs and adjusted for age, sex, calendar period, and purchasing of the other drugs. Purchase of tablets once was chosen as the reference. As can be seen, during all periods with two or more prescriptions of lithium, the rate of dementia was significantly decreased compared with the rate during the period with one prescription of lithium. For all other type of drugs, the rate of dementia during multiple prescription periods did not differ significantly from the rate during prescription period one.

In the multiple Poisson regression analysis, we found no significant interaction between the four exposure variables. This means that the relative increase or decrease in risk attributed to the prescriptions of a specific drug was independent of exposure to other drugs. Nevertheless, all analyses were also undertaken with adjustments for these potential interactions between the four types of drugs. The results only changed marginally in these models, confirming the findings from the model without correction for potential interactions.

In additional analyses, we compared the patients redeeming only one prescription of lithium with those who had redeemed two or more at the end of the study. These two groups did not differ markedly with respect to gender, age at diagnosis, (subtype of) diagnosis, duration of hospitalization, number of previous hospitalizations, previous diagnoses, exposure to antidepressants, exposure to anticonvulsants, or exposure to antipsychotics (results not presented).

Finally, we checked for possible bias due to model misspecification by means of generalized estimating equations using robust standard errors. We repeated the analyses and all estimates, and confidence intervals were reproduced with only minor deviations (results not presented).

Discussion

We found that among patients with mania or bipolar disorder, continued use of lithium was associated with a decreased rate of dementia. In contrast, continued treatment with anticonvulsants, antidepressants, or antipsychotics was not significantly associated with a decreased rate of dementia.

These findings confirm our prior finding among all 16,238 persons prescribed lithium and all 102,644 persons prescribed anticonvulsants during

Table 3. Rate of dementia related to number of prescriptions of lithium, anticonvulsants, antidepressants and antipsychotics, respectively (reference = 1 prescription) and adjusted for age, sex, calendar period and purchasing of the other drugs^a

Number of prescriptions	Lithium RR (95% CI)	Anticonvulsants RR (95% CI)	Antidepressants RR (95% CI)	Antipsychotics RR (95% CI)
1	1	1	1	1
2–4	0.46 (0.21–1.01)	1.49 (0.53–4.19)	1.15 (0.50–2.62)	0.77 (0.48–1.24)
5–9	0.38 (0.16–0.86)	1.81 (0.66–5.01)	0.99 (0.44–2.22)	0.79 (0.47–1.31)
10–19	0.39 (0.19–0.81)	2.24 (0.86–5.81)	0.94 (0.43–2.04)	0.97 (0.59–1.60)
≥ 20	0.44 (0.23–0.85)	1.05 (0.38–2.93)	0.87 (0.42–1.80)	0.80 (0.50–1.30)

RR = rate ratio; CI = confidence interval.

^aPurchasing drugs other than the drug of interest was included in the model as a time-dependent variable. For example, in the analysis of the association between lithium and dementia, the model was adjusted for any purchase of anticonvulsants, antidepressants, or antipsychotics.

the same study period in Denmark (8). In that study, the rate of dementia decreased for persons who continued taking lithium to the rate for the general population, whereas the rate increased with continued use of anticonvulsants. Interestingly, the prevalence of dementia was increased in patients with a diagnosis of mania/bipolar disorder in the present study (71.4/10,000 person-years; Table 1) compared to the prevalence among all users of lithium [62.5/10,000 person-years (8)]. It should be noted that the median age at inclusion was similar in the two studies [52.5 years (quartiles: 45.1–61.6) in the present study (Table 1) versus 52.5 years (quartiles: 43.6–64.8) in our prior study (8)], as was the gender composition (59.2% female versus 63.5%). Patients in the present study all had contact with psychiatric hospital services, whereas patients in our prior study included all patients who were prescribed lithium in Denmark, i.e., from all primary as well as secondary services. Thus, our findings point toward a dose-response relationship between severity of bipolar disorder and the risk of developing dementia: the more severe the disorder, the higher the risk of dementia. In this way, the findings from the two studies confirm prior findings that the risk of developing dementia increases with the number of prior affective episodes (3, 20). The finding in the present study is further in accordance with the findings by Nunes et al. (10) of a decreased prevalence of Alzheimer's disease among patients with chronic lithium therapy, but contrasts with the finding by Dunn et al. (9) of an increased risk of dementia with increased numbers of lithium prescriptions. The latter study compared the number of lithium prescriptions among individuals with and without dementia. As argued by Gattaz et al. (21) and acknowledged by the authors (9), the finding may be due to "reverse causation" related to the sampling method, as affective disorder may increase both the risk of dementia (1, 2) and the likelihood of receiving lithium following a diagnosis of dementia (22). In our prior study (8) as well as in the present study, patients who received a diagnosis of dementia before start of lithium therapy were excluded.

Limitations

Diagnoses were made by clinicians and were not research based. The validity of the ICD-10 diagnoses of bipolar disorder has not been investigated, and indeed, clinicians may have heterogeneous ways of handling diagnoses. However, specialists in psychiatry or doctors training to become specialists in psychiatry made the majority of the diagnoses, and all Danish specialists in psychiatry have

completed identical courses in theoretical and clinical training, which helps to increase the reliability of the diagnoses.

Further, the validity of the ICD-10 diagnosis of dementia in the DNHR and the DPCR has been found to be high, with a correct register diagnosis in 85.8% of the cases as compared to research-based diagnoses (23). The ICD-10 subtype diagnosis of dementia was rather low, but Alzheimer's disease, although under-registered, also had a good validity once the diagnosis was registered. It is estimated that at least two-thirds of patients with dementia in Denmark will get the diagnosis during contact with secondary hospital health care (Phung TK, Waltoft BD, Kessing LV, Mortensen PB, Waldemar G. Is dementia underdiagnosed? A nationwide population-based study of incidence and prevalence of dementia using national registers. Unpublished manuscript, 2008). Nevertheless, it should be noted that general practitioners or private practicing specialists in psychiatry or neurology might diagnose persons with dementia without them being recorded in the registers. In this way, it is most likely that only patients with the more severe forms of dementia are recorded in the registers.

A majority of patients included in the study used drugs from two or more different types of the four drug types, which might reduce the possibility of finding effects related to individual types of drugs. However, all analyses were adjusted for the effect of types of drugs other than the drug of interest. Further, it may be argued that, despite frequent use of polypharmacy, the study revealed very different patterns in the rate of dementia in relation to the use of lithium compared to the use of anticonvulsants/antidepressants/antipsychotics (Table 3). The register contains no data on the prescribed dose of the drugs, but we find no reason to believe that this should have affected our findings in any systematic way.

During the 10-year study period, the use of antipsychotics for bipolar disorder increased, and a total of 80.3% of the sample used antipsychotics at least once. All analyses in the study were adjusted for the effect of calendar period in one-year periods—in this way also taking account of any changes in the health care system during the study period.

Our study did not include sufficient statistical power to analyze the effect of individual anticonvulsants, antidepressants, or antipsychotic drugs.

The most serious concern is the nonrandomized nature of the study, which could be remedied by the use of appropriate propensity scores. In order to make such scores, we would need to predict with

some accuracy the number of drug prescriptions of each patient in the study based on the psychiatric history of the patient combined with other demographic variables. Unfortunately, the exposure of the patients in our data is largely unpredictable, and we have, in fact, access to few relevant demographic or clinical variables for that purpose. We compared the patients redeeming only one prescription of lithium with those who had redeemed two or more at the end of the study. These two groups did not differ markedly with respect to gender, age at diagnosis, (subtype of) diagnosis, duration of hospitalization, number of previous hospitalizations, previous diagnoses, or exposure to antidepressants, anticonvulsants, and antipsychotics.

Interpretation of the results

Our finding of a decreased rate of dementia in relation to continued use of lithium could be the result of a number of potential relationships, as argued in our prior paper on all prescribed lithium in Denmark (8). First, continued treatment with lithium may not be given to patients who the clinician believes will develop dementia later on, e.g., patients with mild cognitive impairment, as clinicians may believe that lithium may induce confusion in such patients. An argument against this explanation is that one should presume that clinical evaluation of cognitive function is undertaken before starting lithium treatment and not when the patient has consumed tablets from several prescriptions of lithium. Additionally, we excluded all patients with a diagnosis of dementia before the start of lithium. Further, continued treatment with lithium may not be given to patients who suffer from cardiac or renal insufficiency that may be related eventually to increased risk of dementia. Clinicians are supposed to check for such somatic illnesses, taking electrocardiograms and blood tests for creatinine/carbamide levels before beginning lithium treatment (24), and similar precautions need to be followed in relation to some anticonvulsants, e.g., carbamazepine. Second, patients who continue taking lithium for long periods may presumably be good compliers with less alcohol use and a healthier lifestyle associated with decreased risk of developing dementia. In relation to both the first and the second explanation, one should presume that the same relationships should occur with the prescription of anticonvulsants, which during the study period was also recommended as prophylactic long-term treatment for bipolar disorder. Anticonvulsants may

be associated with confusion among patients with mild cognitive impairment much in the same way as lithium (25), and taking anticonvulsants for longer periods requires good compliance possibly related to a healthier lifestyle. Finally, anticonvulsants have been associated with neuroprotective abilities in much the same way as lithium, among others, including inhibition of glycogen synthase kinase-3 (26, 27).

At first glance, the early decrease in the rate of dementia following consumption of lithium tablets corresponding to one purchase is somewhat spurious. According to our results, a protective effect of lithium seems to start following intake of a few hundred tablets of lithium. However, an increase in human brain gray matter (28–30) and in N-acetyl-aspartate levels [a putative marker of neuronal viability and function (31)] have been found following only four weeks of lithium treatment. Additionally, no correlation has been found between length of lithium treatment and gray matter brain volume (29).

It should be recalled that data from our prior studies suggest that the rate of dementia increases with the number of affective episodes in patients with bipolar (and unipolar) disorder (3). Thus, according to those findings, the rate of dementia should increase with time and with the number of prescriptions in the present study. However, this was not the case; the rate of dementia stayed low during continued lithium use, more than halving the rate during prescription period one. Thus, although we do not find it likely that our results can be explained fully by methodological drawbacks, we cannot rule out this possibility due to the nonrandomized nature of our data.

It is not possible from the present study to determine whether lithium has a protective effect against dementia due to increased neurogenesis or due to mood-stabilizing abilities preventing recurrence of affective episodes or other treatment-related factors. In naturalistic data, as used in the present study, it is not possible validly to investigate the association between treatment, such as lithium, and the course of illness, due to confounding by indication, as patients with many episodes tend to take lithium for a longer time (32).

Our present findings regarding antidepressants did not confirm our prior finding among all users of antidepressants in Denmark of a decreased rate of dementia in relation to continued use of antidepressants (18). It is possible that continued use of antidepressants might be associated with a decreased rate of dementia in patients with depressive (unipolar) disorder and anxiety disorders only and not in patients with bipolar disorder. No other

study has been published on the effect of antidepressants in unipolar or bipolar disorder.

Similarly, no other study has, to the best of our knowledge, investigated the effect of antipsychotics on the risk of developing dementia. We did not find that continued use of antipsychotics was associated with a reduced rate of dementia, in contrast to suggestions of improvement of cognition in patients with schizophrenia treated with antipsychotics (33–36; reviewed by Grada and Dinan in 37).

Conclusions

Among patients with mania or bipolar disorder, continued use of lithium was associated with a decreased rate of dementia. The rate of dementia decreased early following consumption of tablets corresponding to one prescription of lithium. In contrast, no significant associations were found between continued treatment with anticonvulsants, antidepressants, or antipsychotics, respectively, and dementia. Methodological reasons for these findings cannot be excluded due to the nonrandomized nature of data.

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